Dataverse in the Mesh

Interoperating with a Data Mesh — The Perspective from Data Repositories

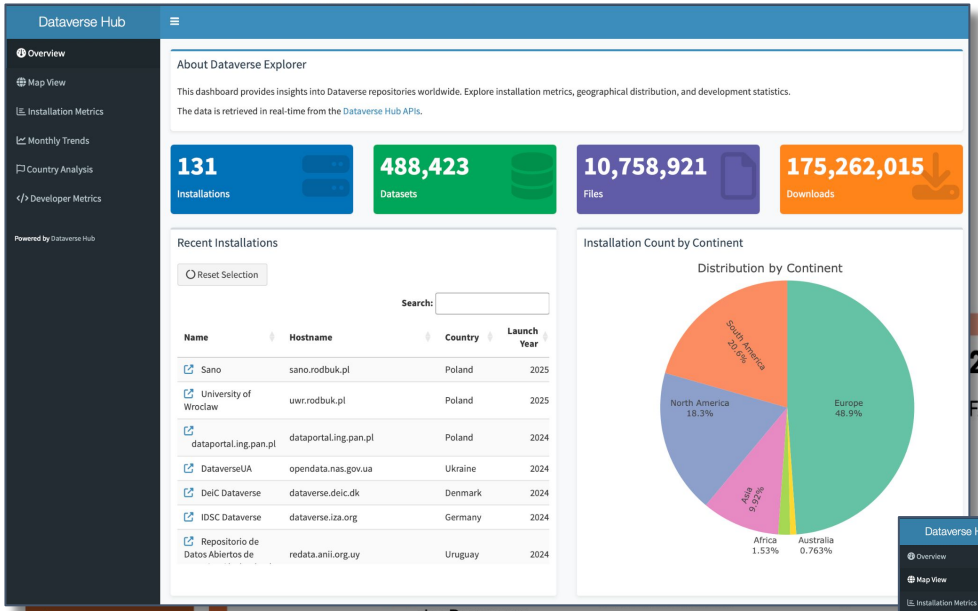
Stefano Iacus, Sonia Barbosa, James Myers











Wilkinson, et al. (2016). The FAIR guiding principles for scientific data management and stewardship. Scientific Data, 3.

2016

FAIR data

2023

100th Dataverse installation

Dataverse 6.0

Dataverse Re-architecture

2024

The Next Frontier!

Large data

Computing on data

10th DVCM

2025

131 Installations!

New SPA

Dataverse HUB

MarketPlace

HPC - DV Integration

External Search

Large Data Services

ML + MCP

of new AI stuff!

gration

1997

Project Begins

Focus on preserving & sharing social sciences data

Harvard-MIT Data Center collaboration

Murray Data Archive

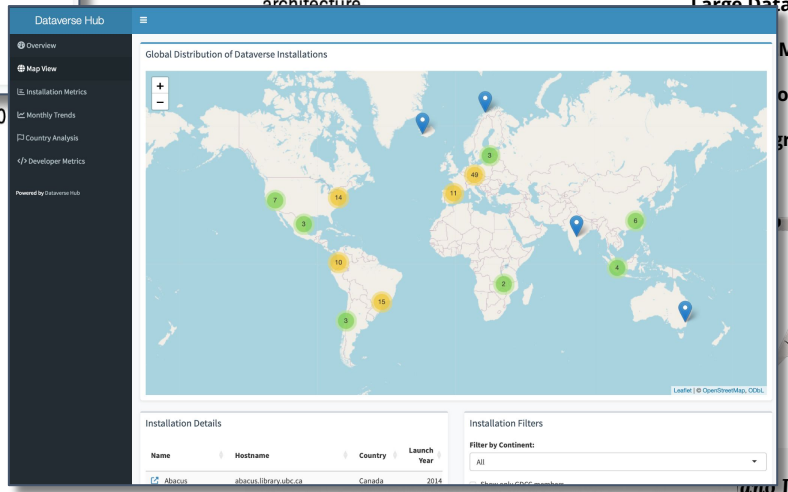
The Dataverse Network 1.0

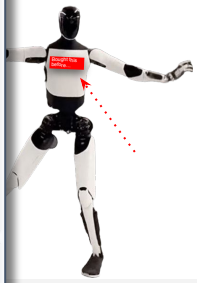
Harvard Dataverse Repository established

the Dataverse Network

Dataverse 4.0

Altman, Borgman, Crosas & Matone. (2015). An introduction to the joint principles for data citation. Bulletin of the Association for Information Science and Technology, 41(3), 43–45.





https://askdataverse.shinyapps.io/dvhub/

Slavko Vasiljevic, Vera Iacus, James Myers

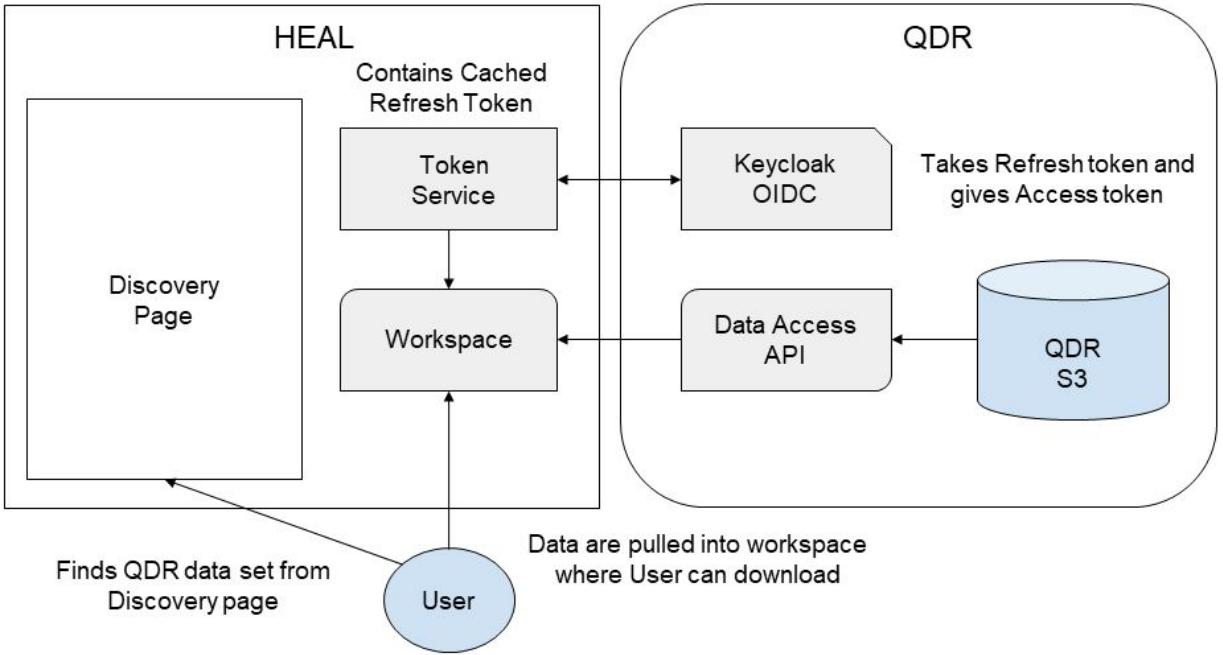
Driving Mesh Use Cases

- NIH Helping to End Addiction Long-term (HEAL) Initiative (QDR, Harvard, …)
- Harvard Data Commons/ Massachusetts Open Cloud/NESE Integration
- Norwegian Research Commons/ DataverseNO 2030
- Australian Data Archive access request review process
- RSpace integration using ISGNs
- Local Contexts/TKLabels
- React-based Dataverse front end, third-party tools
- Common theme: many-to-many integrations based on standards
- …

“Pre-Mesh” Integration Capabilities

- PIDs
  - DOIs, Handles for datasets & files, “PermaLinks” for catalog use cases
  - ORCIDs, ROR, Licenses, Controlled vocabularies
  - Semantic Mapings
- Configurable Metadata Blocks, Standards-based Exporters
  - Native JSON, DDI, RDA-conformant OAI-ORE, DataCite/OpenAire, …
- External Tools
  - Ability to launch viewers, notebooks, metadata input, other tools
  - OIDC, Signed URLs, API keys for controlled API access
- Storage/Transfer integration
  - Files, S3, Globus, DropBox, Github, iRods …

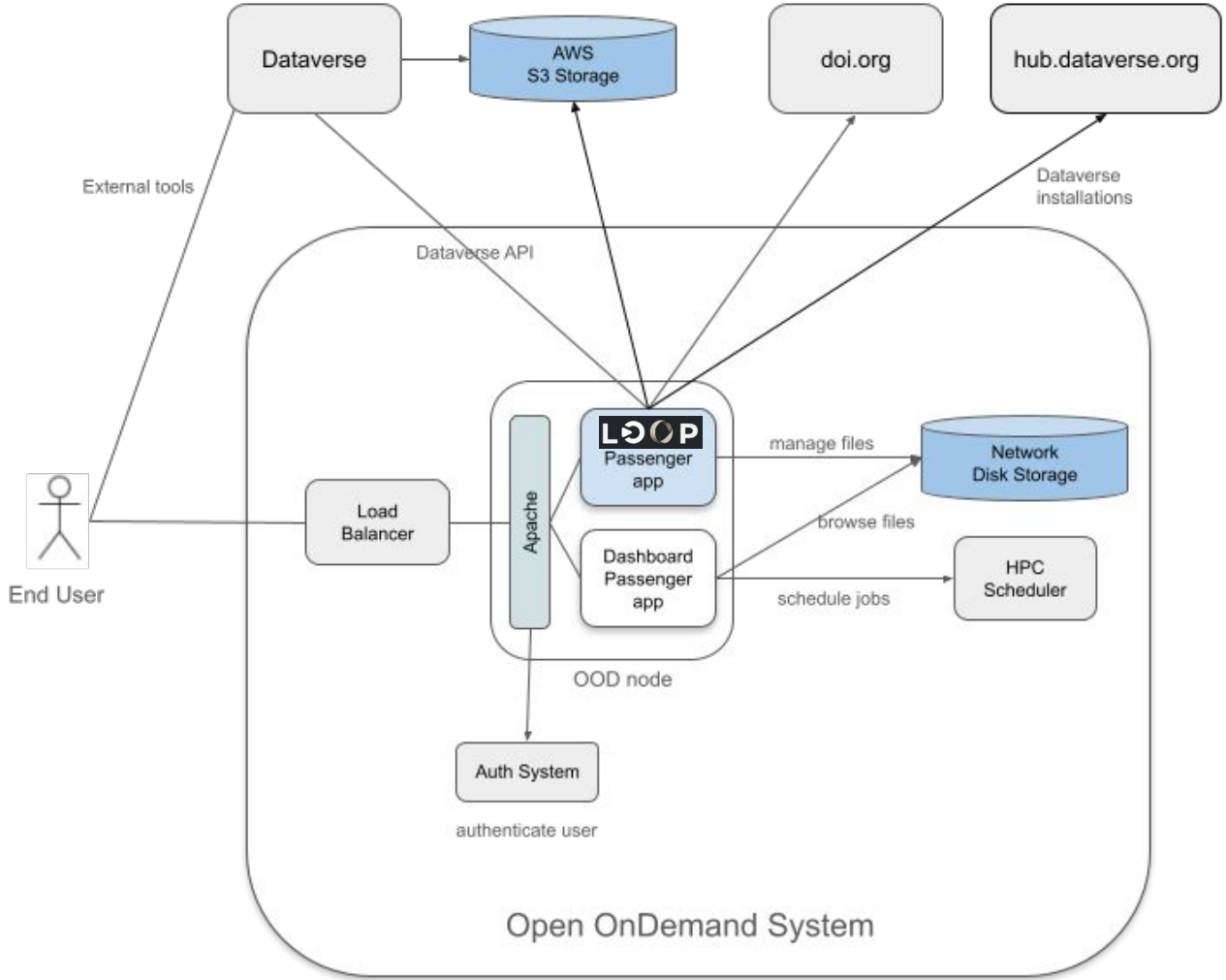
Populating a HEAL Workspace with data from Dataverse



OIDC Linking of Accounts

Access control enforced by QDR API

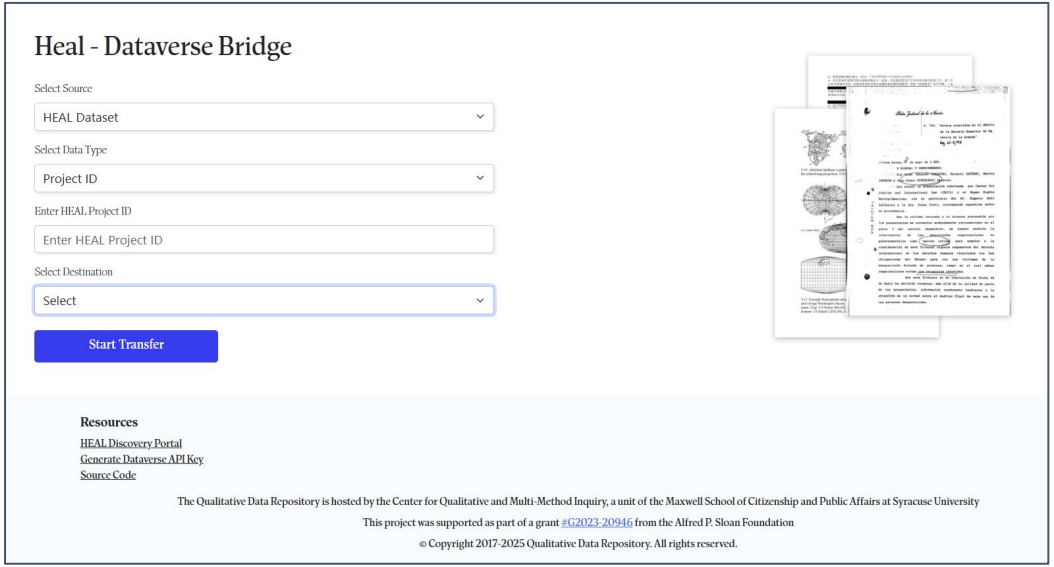
Diagram courtesy of Brienna Larrick, Sr. Scientific Project Manager for HEAL


Dataverse
AWS
S3 Storage
doi.org
hub.dataverse.org
External tools
Dataverse
installations
Dataverse API
LOOP
Passenger
app
manage files
Network
Disk Storage
Load
Balancer
Apache
browse files
Dashboard
Passenger
app
HPC
Scheduler
schedule jobs
End User
OOD node
Auth System
authenticate user
Open OnDemand System

Creating a Dataverse Dataset based on a Registered HEAL project

Lightweight app to map HEAL metadata to Dataverse standard and custom HEAL metadata

- Select HEAL ID,
- Create Dataset



Created by Kamaljit Kaur@Syr

Try it at :

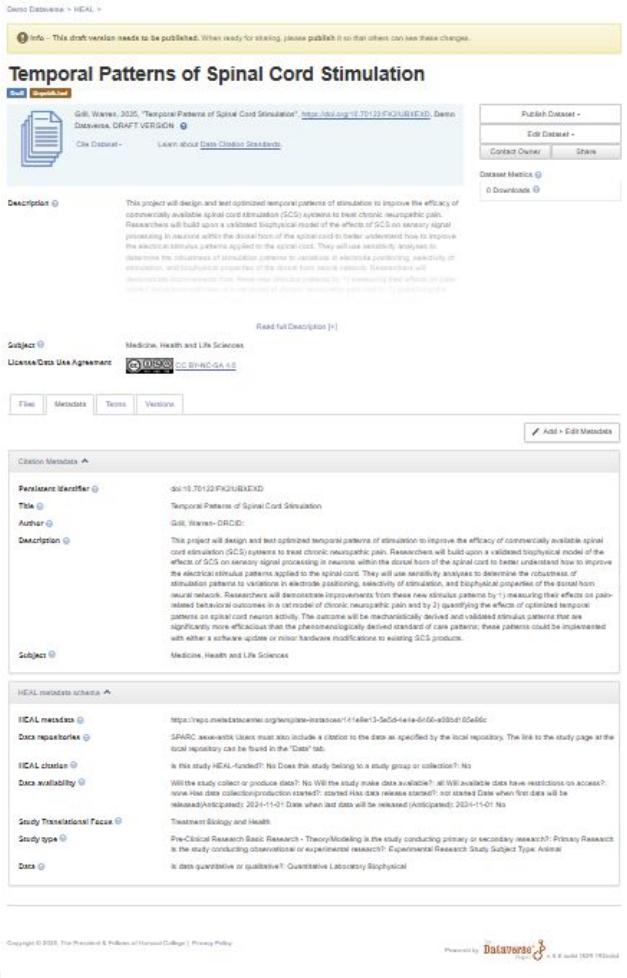
https://demo.dataverse.org/dataverse/heal/

Draft Dataset

All metadata transferred

Ready to add files

Publication makes data available from Dataverse and back in the mesh



Data Advertised to Meshes

Signposting - A standard machine readable way to find metadata and download URLs

Croissant - Extension of Schema.org exposing variable-level metadata for machine learning

Dataverse Model Context Protocol server - exposing Dataverse capabilities to AI Agents

Dataverse in the Mesh

- Building on Mesh Design Principles and Community Standards
- Advertising its Capabilities in Machine-actionable ways

Thanks for your attention!